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Delivering applications to client machines in third-party environments

Abstract

A computerized method efficiently and instantaneously delivers an application in a third-party environment. An application plugin associated with the application is published to an application publishing portal, while the application itself is stored in an application repository. The application plugin is a fraction of a size of the application. A request is received from a client machine to download the application. In response to the request, the application plugin, instead of the application, is provided to the client machine. The application plugin, comprising metadata and instructions to locate the application on the application repository, is used to make the application available to the user without installing the application on the client machine. The disclosure enables savings on network bandwidth, memory usage, and processing when delivering the application to the client machine.

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Background/Summary

BACKGROUND

(1) Existing application delivery environments include an application publishing portal which stores a complete application package for delivery to end users. To install an application on the devices of the end users, the complete application package is pushed to each of the devices. This causes large downloads to be replicated on every single user device needing the application, resulting in slow application availability after user login. Because applications are often large in size (e.g., a few gigabytes), application delivery in such environments consumes significant computing resources including bandwidth and storage. For example, after delivery, a separate copy of the application exists on each of the devices.

(2) Further, many application delivery environments that publish applications for delivery to an end user do not provide an application management console to facilitate operability with different application managers and different application formats.

SUMMARY

(3) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(4) In some examples, a computerized method for efficiently and instantaneously (e.g., in almost real-time) delivering applications to client machines in a third-party application publishing environment is provided. An application manager on a client machine sends a request to download

an application from an application publishing portal. In response, the application publishing portal provides an application plugin associated with the application. The application itself is stored in an application repository. The application plugin is a fraction of a size of the actual application, and comprises metadata and instructions to locate the application on the application repository. On receipt of the application plugin associated with the application, one or more of the metadata and the instructions in the application plugin are used to make the application available to the user using on-demand delivery (e.g., virtually, without downloading the application to the client machine). Thus, examples of the disclosure enable savings on network bandwidth, memory usage, and processing associated with accessing the application on the client machine.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present description will be better understood from the following detailed description read considering the accompanying drawings, wherein:
- (2) FIG. 1 illustrates an example architecture that advantageously delivers applications in third-party application publishing environments;
- (3) FIG. 2 is a data flow diagram illustrating example operations for delivering applications in a third-party environment, such as that of FIG. 1;
- (4) FIG. 3A illustrates application packaging in an example architecture, such as that of FIG. 1;
- (5) FIG. 3B illustrates application delivery to client machines in an example architecture, such as that of FIG. 1;
- (6) FIG. 4A is a flowchart illustrating an example method for receiving applications in a third-party environment, such as that of FIG. 1;
- (7) FIG. 4B is a flowchart illustrating an example method for delivering applications in a third-party environment, such as that of FIG. 1;
- (8) FIG. 5 illustrates an example virtualization architecture that may be used as a computing platform; and
- (9) FIG. 6 illustrates a block diagram of an example computing apparatus that may be used as a component of an example architecture, such as the architectures of FIG. 1 and FIG. 5.
- (10) Corresponding reference characters indicate corresponding parts throughout the drawings. In FIGS. 1 to 6, the systems are illustrated as schematic drawings. The drawings may not be to scale. Any of the figures may be combined into a single example or embodiment.

DETAILED DESCRIPTION

(11) In existing systems, an application publishing portal (e.g., Google Workspace, Microsoft Intune, and the like) is used to deliver a full application to a client machine, and the client machine installs the full application received from the application publishing portal. An application manager (e.g., VMware App Volumes) may not have management control over a particular application publishing portal (e.g., a third-party application publishing portal). In such a third-party environment, the application manager and the application publishing portal are different entities or parties. Aspects of the disclosure enable a fast and efficient way to deliver applications to client machines by downloading an application plugin from a third-party application publishing portal instead of the application itself.

(12) The application plugin, for the application, is created by an application packager in a format supported or otherwise accepted by the application publishing portal. The application plugin is a collection of metadata and instructions related to the application (but without the actual application). While the application plugin is packaged in a format that the third-party application publishing portal uses for publishing applications, the application itself is stored in the application repository. For example, if the format required by the application publishing portal is an .msi

format or .intunewin format, the application packager of the disclosure creates an application plugin in that required format. The application plugin is very lightweight and small in size, as it is not the application itself. The application plugin is accepted and published on the application publishing portal as if it were a regular, full application, because the application plugin appears to be a regular, full application to the application publishing portal. Further, the application plugin is assigned to particular users just like a regular application is assigned to users on the application publishing portal. The metadata and instructions are designed in a way that when the application plugin is processed on the client machine (e.g., an end user machine), the application plugin makes the regular, full application available to the client machine from the application repository (e.g., not the application publishing portal) where the regular, full application is actually stored. In some examples, the application repository may be a third-party application repository that is managed by an entity or party different from the application publishing portal and/or the application manager.

(13) The disclosure advantageously provides access to an application without actually delivering and installing the application from the application publishing portal to the client machines. Instead of the application (e.g., having a memory size of a few gigabytes (GBs)), an application plugin (e.g., having a memory size of a few kilobytes (KBs)) is delivered from the application publishing portal and processed on the client machines. The processing of the application plugin includes using the metadata and/or instructions in the application plugin to find the application on the application repository. The application is then made available to the client machine. For example, the client machine is a virtual machine, and the application is made available on the client machine virtually, on-demand. An example of on-demand delivery of an application to a client machine is VMware's App Volumes, which provides dynamic mounting of a virtual disk volume on the client machine to make the application appear as if was installed on the client machine.

(14) In this way, aspects of the disclosure reduce consumption of (1) network bandwidth because a smaller footprint application plugin is downloaded instead of the application, and (2) an amount of memory used for making the application available on the user machines because the application is not installed in the traditional sense. This improves the functioning of the underlying computing device.

(15) FIG. 1 is a block diagram illustrating an example system **100** that advantageously delivers applications in third-party environments. In some examples, the system **100** includes an application publishing portal **102** on which applications from developers are published. The application publishing portal **102** may be under control of a company which facilitates publication of applications from developers on the application publishing portal **102**. Aspects of the disclosure submit an application plugin **104** for publication on the application publishing portal **102** just like a regular, full application is published on the application publishing portal **102**. The application plugin **104** may be published in one or more formats supported by the application publishing portal **102**. For example, the format of the application plugin **104** may be .intunewin or .msi, as supported by the application publishing portal **102**.

(16) Separately, the application is stored in an application repository **106**. For example, the application repository **106** may store applications in different formats (e.g., APP1 **108** in VHD format, APP2 **110** in MSIX format, APP3 **112** in VHDX format and the like). The application repository **106** may store one application in multiple, different formats (e.g., APP4 in VHD format, APP4 in VHDX format, or on any other format). In some examples, the application publishing portal and the application repository may be on different storage devices or on the same storage device.

(17) In some examples, the request to access an application is automatically generated by the client machine upon detecting a login into the client machine by a user, because the application has been assigned to the user (e.g., associated with an access profile of the user). For example, an application manager **118** on a client machine **116** sends a request to the application publishing portal **102** to download the application (such as application **108**, **110**, or **112**). However, the application

publishing portal **102** stores the application plugin **104**, and provides the application plugin **104** to the client machine **116** in response to the request. The application plugin **104** is a fraction of a size of the application. For example, the application plugin is of the order of a few KBs (e.g., around 10 KB or 20 KB) whereas the application can be of the order of a few GBs (e.g., 1 GB, 2 GB, 3 GB, or even more).

(18) As such, in response to the request to download the application, the application plugin **104** associated with the application is received from the application publishing portal **102** by the client machine **116**. The application plugin **104** comprises metadata and instructions to locate the application (such as application **108**, **110**, or **112**) on the application repository **106**. Using one or more of the metadata and the instructions, the application associated with the application plugin **104** is made available to the user **114**. For example, the application is provided virtually, on-demand by mounting a virtual disk to give the appearance of installation of the application, but without actually installing the application on the client machine **116**. The client machine **116** may be a virtual machine (e.g., a VM **501-508** of FIG. 5) or a computing device such as a computing apparatus **618** of FIG. 6.

(19) FIG. 2 illustrates a data flow diagram **200** with example operations for delivering applications in third-party environments in an example architecture such as that of FIG. 1. In Operation 1, an application plugin (such as application plugin **104**) is uploaded to the application publishing portal **102**. The application plugin **104**, or the application associated with the application plugin, is assigned to one or more users at the application publishing portal **102** in Operation 2. The application is stored in the application repository **106** in Operation 3. For example, the applications stored in the application repository **106** may be in different formats (such as applications **108-112**). When the user **114** logs into the client machine **116** in Operation 4, software, hardware, or firmware (such as the application manager **118**) detects that the application (or application plugin **104**) has been assigned to the user **114**. In some examples, the software may periodically detect if any additional application or application plugin **104** has been assigned to the user **114**.

(20) In Operation 5, the client machine **116** sends a request to the application publishing portal **102** for the application (e.g., the application plugin **104**) that has been assigned to the user **114**. In Operation 6, the application plugin **104** is downloaded from the application publishing portal **102** to the client machine **116**. The application plugin **104** is processed on the client machine in Operation 7. In Operation 8, the application plugin **104** including metadata and instructions locates the application in the application repository **106** (e.g., with or without the help of the application manager). The application is then made available to the user **114** on the client machine **116**, such as by the application manager.

(21) In some examples, instead of locating the application in the application repository **106**, the application plugin **104** causes the application manager to locate the application on the application repository **106** using the metadata and/or instructions. For example, the application plugin **104** makes a procedure call to the application manager, and conveys the metadata and/or instructions for use by the application manager. If the application plugin **104** is not able to find the application manager, the instructions in the application plugin may be used to find the application manager from a source (such as the application publishing portal or some other source) and install the application manager on the client machine **116**. Thereafter, the application manager, using the metadata and/or the instructions in the application plugin, locates the application on the application repository **106** and makes the application available to the user **114** on the client machine **116**.

(22) FIG. 3A illustrates an example of application packaging used in an example architecture, such as the architecture of FIG. 1. In existing systems, an application packager packages an application for download by creating a regular application package (e.g., in VHD format) for publishing on the application publishing portal. In contrast, in aspects of the disclosure, the application packager packages an application for download by creating a regular application package (e.g., SpecificApp.VHD **308** of around 3 GB size), but also creates an application plugin (e.g.,

SmartPlugin **304** of around 20 Kb size) associated with the regular application package. In this example, the SpecificApp.VHD **308** is then stored in the application repository **106**, while the SmartPlugin **304** is published on the application publishing portal **102**. Thus, the SmartPlugin **304**, rather than the SpecificApp.VHD **308**, is published on the application publishing portal **102** just like a regular application and assigned to users.

(23) In creating the SmartPlugin **304**, the application packager includes metadata and/or instructions for obtaining the SpecificApp.VHD **308**. For example, the metadata describes SpecificApp.VHD **308** and includes a location of SpecificApp.VHD **308** in the application repository **106**. The metadata may be in a format such as JavaScript Object Notation (JSON), Extensible Markup Language (XML), etc. The instructions in the SmartPlugin **304** comprise executable instructions that trigger execution of the application manager **118** on the client machine **116**. For example, the instructions in the SmartPlugin **304** comprise executable instructions that trigger a remote procedure call (RPC) end point in the application manager **118**. In operation, the application manager **118** selects, based on the RPC end point, an application format of SpecificApp.VHD **308** from the application repository **106** depending on an execution environment of the SmartPlugin **304** on the client machine **116**.

(24) In some examples, one application plugin **104** may be associated with one application. Alternatively, one application plugin **104** may be associated with a suite of applications. Another alternative is that many applications plugins **104** may be associated with a single application (e.g., each application plugin **104** being intended for a particular execution environment of the client machine), or suite of applications.

(25) Referring next to FIG. **3B**, application delivery to multiple client machines is illustrated. An application packager creates an application plugin (e.g., SmartPlugin **304**) for an application (e.g., SpecificApp.VHD **308**). The SmartPlugin **304** is published on a third-party application publishing portal **302** and the SpecificApp.VHD **308** is stored on an application repository (e.g., third-party file share **306**). The third-party file share **306** may or may not be under control of the entity that controls the third-party application publishing portal **302** (e.g., the third-party application publishing portal **302** may be managed by an entity that is different than an entity managing the application manager on the client machine).

(26) The SmartPlugin **304** may be assigned to users (e.g., such as users **314A-314C**). The number of users to whom the SmartPlugin **304** is assigned may be in the hundreds, thousands, or even more. When these users log in into their client machines (e.g., client machines **316A-316C**), only the SmartPlugin **304** is downloaded from the third-party application publishing portal **302** and processed on each client machine **316A-316C**. The SmartPlugin **304** may be used, directly or with the help of application manager, to access the SpecificApp.VHD **308** stored in the third-party file share **306**. The SpecificApp.VHD **308** is then made available to the users **314A-314C** without actually downloading the whole application on each client machine **316A-316C**. This is accomplished using, for example, virtual on-demand delivery of SpecificApp.VHD **308**.

(27) The example of FIG. **3B** illustrates how multiple machines access the same application on the application repository **106** yet avoid the need to download the full application on every single machine. This generates significant savings in both storage and time required for application delivery.

(28) FIG. **4A** is a flowchart illustrating an example method **400A** for delivering applications in third-party environments in the architecture of FIG. **1**. At **402**, an application manager, such as application manager **118** on a client machine **116**, sends a request to download an application from an application publishing portal that stores an application plugin associated with the application. The application plugin is a fraction of a size of the associated application. The application itself is stored in an application repository by an application packager. At **404**, in response to the request to download the application, the application plugin **104** associated with the application is received from the application publishing portal **102**. At **406**, the application plugin **104** is processed on the

client machine **116** by the application manager **118**. The application plugin **104** comprises metadata and instructions to locate the application on the application repository. At **408**, the application is made available to the user **114** without installing the application on the client machine **116**. For example, the application is made available to the user **114** without downloading the application to the client machine **116**.

(29) FIG. **4B** is a flowchart illustrating an example method **400B** for delivering applications in third-party environments. At **410**, an application plugin (e.g., application plugin **104**) associated with an application is published to an application publishing portal (e.g., application publishing portal **102**). The application plugin **104** is assigned to a user (e.g., user **114**) at **412** and the application is stored in an application repository (e.g., application repository **106**) at **414**.

(30) In some examples, the application plugin **104**, which includes metadata and instructions, is in a format that the application publishing portal **102** accepts. The application plugin **104** may be created, by the application packager, to correspond to any application format or type, making it possible to publish different application formats using the application plugin **104** as long as the application format can be identified and used by a client machine (e.g., client machine **116**). For example, the format of application plugin **104** may be .vhd, .vhdx, .msi, .msix, “app-v”, “pin-app”, etc.

(31) At **416**, a request to download the application is received from the client machine **116**. In response to the request to download the application, the application plugin **104** is provided to the client machine **116** at **418**. The application plugin **104** encapsulates the metadata and/or instructions. The metadata in the application plugin **104** describes the application package including, for example, application package identifier, application VHD name, relative location of the application on the file share (e.g., the application repository **106**), etc. The instructions part of the application plugin **104** includes, for example, computer-executable instructions that when executed by a processor trigger execution of an application manager (e.g., application manager **118**). In some examples, the instructions are in the form of an RPC call exposed by the application manager **118** executable for handling such a functionality. The RPC endpoint may also accept the metadata related to the application.

(32) At **420**, the application is located on the application repository **106** using the application plugin **104**. For example, the application manager **118** uses the metadata in the application plugin **104** to know that the application (e.g., the .vhd file of the application) is located at the application repository **106**, as well as where the application is located in the application repository **106** (e.g., a relative path). An application may exist in multiple package formats (e.g., .vhd, .vhdx, .msi, .msix, etc.). Applications may also be in common formats like ThinApp, Microsoft MSIX, App-V etc. or any formats. When delivering other application formats, the application plugin **104** may have different metadata to describe the application. In some use cases, the instructions encapsulated in the application plugin might also trigger an RPC end point in the application manager that lets the application manager decide what application format to choose from the application repository **106** depending on the environment it is running in. In some examples, this is done by providing details identifying the application (e.g., application globally unique identifier, name, etc.) but not specifying the format type. The communication between application plugin and application manager may also be done using mechanisms other than an RPC call into the process.

(33) In some examples, there are additional RPC endpoints exposed by the application manager **118** components (e.g., svservice) to facilitate making the application available without installation of the application.

(34) The application manager **118** then processes all this information passed into the RPC endpoint, and makes the application available from the application repository **106** at **422**. The application manager **118** makes the application available on the client machine **116** without downloading the application from either the application publishing portal **102** or the application repository **106**. Rather, the application is made available virtually on-demand (e.g., by mounting a virtual disk).

(35) In some examples, the instructions in the application plugin **104** may call the application manager **118** which is then run to virtualize the application on the client machine **116**. In some other examples, if the application manager **118** is not available on the client machine **116**, the application plugin **104** uses the metadata and/or instructions therein to download and install the application manager **118** relevant to the execution environment of the client machine **116**. Thereafter, the application manager **118** may use the application plugin **104** to virtualize the application on the client machine **116**. In some other examples, if the application manager **118** is not available on the client machine **116**, the metadata and/or instructions in the application plugin **104** may be directly used to virtualize the application on the client machine **116** without requiring the application manager **118**.

(36) In some cases, instead of virtually delivering the application, the application is completely installed on the machine. In such instances, the application plugin **104** works with the application manager **118** to install the application on the client machine **116**. For example, the application plugin **104** finds the application installer format (e.g., .msi etc.) on the application repository **106** in situations where an install of the application on the client machine **116** is needed, and installs the application on the client machine **116**. The decision to install the application on the client machine **116** may be made at runtime depending on the environment the application plugin **104** is delivered in.

(37) Examples of architecture **100** are operable with virtualized and non-virtualized storage solutions. FIG. 5 illustrates a virtualization architecture **500** that may be used. Virtualization architecture **500** is comprised of a set of compute nodes **521-523**, interconnected with each other and a set of storage nodes **541-543** according to an embodiment. In other examples, a different number of compute nodes and storage nodes may be used. Each compute node hosts multiple objects, which may be virtual machines (VMs, such as base objects, linked clones, and independent clones), containers, applications, or any compute entity (e.g., computing instance or virtualized computing instance) that consumes storage, such as local storage **561, 562, 563** (e.g., storage devices directly attached to the compute nodes **521, 522, 523**) or storage provided by other devices or services (e.g., SAN storage, network-attached storage (NAS), cloud storage, storage provided by storage nodes **541, 542, 543**, or the like). When objects are created, they may be designated as global or local, and the designation is stored in an attribute. For example, compute node **521** hosts objects **501, 502**, and **503**; compute node **522** hosts objects **504, 505**, and **506**; and compute node **523** hosts objects **507** and **508**. Some of objects **501-508** may be local objects. In some examples, a single compute node may host **50, 100**, or a different number of objects. Each object uses a VM disk (VMDK), for example VMDKs **511-518** for each of objects **501-508**, respectively. Other implementations using different formats are also possible. A virtualization platform **530**, which includes hypervisor functionality at one or more of compute nodes **521, 522**, and **523**, manages objects **501-508**. In some examples, various components of virtualization architecture **500**, for example compute nodes **521, 522**, and **523**, and storage nodes **541, 542**, and **543** are implemented using one or more computing apparatus such as computing apparatus **618** of FIG. 6.

(38) Virtualization software that provides software-defined storage (SDS), by pooling storage nodes across a cluster, creates a distributed, shared data store, for example a storage area network (SAN). Thus, objects **501-508** may be virtual SAN (vSAN) objects. In some distributed arrangements, servers are distinguished as compute nodes (e.g., compute nodes **521, 522**, and **523**) and storage nodes (e.g., storage nodes **541, 542**, and **543**). Although a storage node may attach a large number of storage devices (e.g., flash, solid state drives (SSDs), non-volatile memory express (NVMe), Persistent Memory (PMEM), quad-level cell (QLC)) processing power may be limited beyond the ability to handle input/output (I/O) traffic. Storage nodes **541-543** each include multiple physical storage components, which may include flash, SSD, NVMe, PMEM, and QLC storage solutions. For example, storage node **541** has storage **551, 552, 552**, and **553**; storage node **542** has storage **555** and **556**; and storage node **543** has storage **557** and **558**. In some examples, a single

storage node may include a different number of physical storage components.

(39) Storage nodes **541-543** are treated as a SAN with a single global object, enabling any of objects **501-508** to write to and read from any of storage **551-558** using a virtual SAN component **532**. Virtual SAN component **532** executes in compute nodes **521-523**. Using the disclosure, compute nodes **521-523** are able to operate with a wide range of storage options. In some examples, compute nodes **521-523** each include a manifestation of virtualization platform **530** and virtual SAN component **532**. Virtualization platform **530** manages the generating, operations, and clean-up of objects **501** and **502**. Virtual SAN component **532** permits objects **501** and **502** to write incoming data from object **501** and incoming data from object **502** to storage nodes **541**, **542**, and/or **543**, in part, by virtualizing the physical storage components of the storage nodes.

Additional Examples

(40) In some examples, the application plugin is used to make the application unavailable to the user on the client machine. That is, processing the application plugin results in disabling execution of the application on the client machine, or otherwise removing access to the application (e.g., making the application no longer available on-demand).

(41) An example computerized method for delivering an application in a third-party environment comprises: sending, by an application manager on a client machine to an application publishing portal, a request to download the application from the application publishing portal, wherein the application publishing portal publishes an application plugin associated with the application, wherein the application plugin is a fraction of a size of the application, wherein the application is stored in an application repository by an application packager; in response to the request to download the application, receiving, from the application publishing portal, the application plugin associated with the application; processing, by the application manager, the application plugin on the client machine, the application plugin comprising metadata and instructions to locate the application on the application repository; and using one or more of the metadata and the instructions, making the application available to the user without installing the application on the client machine.

(42) An example computer system for delivering an application in a third-party environment comprises: a processor; and a non-transitory computer readable medium having stored thereon program code executable by the processor, the program code causing the processor to: publish an application plugin associated with the application to an application publishing portal associated with the third-party environment, wherein the application plugin is a fraction of a size of the application; assign the application plugin to a user; store the application in an application repository; receive, from a client machine, a request to download the application; and in response to the request to download the application, provide the application plugin to the client machine, the application plugin comprising metadata and instructions to locate the application on the application repository and make the application available to the user without installing the application on the client machine.

(43) An example non-transitory computer storage medium has stored thereon program code executable by a processor, the program code embodying a program code method for delivering an application in a third-party environment comprising: publishing an application plugin associated with the application to an application publishing portal associated with the third-party environment, wherein the application plugin is a fraction of a size of the application; assigning the application plugin to a user; storing the application in an application repository; receiving, from a client machine, a request to download the application; and in response to the request to download the application, providing the application plugin to the client machine, the application plugin comprising metadata and instructions to locate the application on the application repository and make the application available to the user without installing the application on the client machine.

(44) Alternatively, or in addition to the other examples described herein, examples include any combination of the following: the application packager publishes the application plugin on the

application publishing portal, the method further comprising: creating, by the application packager, the application plugin in a format supported by the application publishing portal. the application plugin is automatically downloaded on the client machine upon detecting a login into the client machine by the user. the application is available to the user of the client machine in real time upon detecting the login into the client machine by the user. the application plugin makes a procedure call to the application manager, wherein the application manager locates the application on the application repository and makes the application available from the application repository to the client machine. determining, by the application plugin, that the application manager is not available on the client machine; and based on the determining, making, by the application plugin, the application available on the client machine from the application repository creating, by the application packager, the application plugin for the application. the metadata describes the application and includes a location of the application in the application repository. the metadata is in JavaScript Object Notation (JSON) format or Extensible Markup Language (XML) format. the instructions in the application plugin comprise executable instructions that trigger execution of the application manager on the client machine. the instructions in the application plugin comprise executable instructions that trigger a remote procedure call (RPC) end point in the application manager, wherein the application manager selects, based on the RPC end point, an application format from the application repository depending on an execution environment of the application plugin on the client machine. the application plugin is used to make the application unavailable to the user on the client machine. the request for the application plugin is automatically generated by the client machine upon detecting a login into the client machine by the user. the application is available to the user of the client machine in real time upon detecting the login into the client machine by the user.

Exemplary Operating Environment

(45) The present disclosure is operable with a computing device (computing apparatus) according to an embodiment shown as a functional block diagram **600** in FIG. **6**. FIG. **6** illustrates a block diagram of an example computing apparatus that may be used as a component of the architectures of FIG. **1** and FIG. **5**. In an embodiment, components of a computing apparatus **618** are implemented as part of an electronic device according to one or more embodiments described in this specification. The computing apparatus **618** comprises one or more processors **619** which may be microprocessors, controllers, or any other suitable type of processors for processing computer executable instructions to control the operation of the electronic device. Alternatively, or in addition, the processor **619** is any technology capable of executing logic or instructions, such as a hardcoded machine. Platform software comprising an operating system **620** or any other suitable platform software may be provided on the computing apparatus **618** to enable application software **621** to be executed on the device. According to an embodiment, the operations described herein may be accomplished by software, hardware, and/or firmware.

(46) Computer executable instructions may be provided using any computer-readable medium (e.g., any non-transitory computer storage medium) or media that are accessible by the computing apparatus **618**. Computer-readable media may include, for example, computer storage media such as a memory **622** and communications media. Computer storage media, such as a memory **622**, include volatile and non-volatile, removable, and non-removable media implemented in any method or technology for storage of information such as computer readable instructions, data structures, program modules or the like. Computer storage media include, but are not limited to, hard disks, RAM, ROM, EPROM, EEPROM, NVMe devices, persistent memory, phase change memory, flash memory or other memory technology, compact disc (CD, CD-ROM), digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage, shingled disk storage or other magnetic storage devices, or any other non-transmission medium (e., non-transitory) that can be used to store information for access by a computing apparatus. In contrast, communication media may embody computer readable instructions, data

structures, program modules, or the like in a modulated data signal, such as a carrier wave, or other transport mechanism. As defined herein, a computer storage medium or media does not include communication media. Therefore, a computer storage medium or media does not include propagating signals. Propagated signals are not examples of computer storage media. Although the computer storage medium (the memory **622**) is shown within the computing apparatus **618**, it will be appreciated by a person skilled in the art, that the storage may be distributed or located remotely and accessed via a network or other communication link (e.g., using a communication interface **623**). Computer storage media are tangible, non-transitory, and are mutually exclusive to communication media.

(47) The computing apparatus **618** may comprise an input/output controller **624** configured to output information to one or more output devices **625**, for example a display or a speaker, which may be separate from or integral to the electronic device. The input/output controller **624** may also be configured to receive and process an input from one or more input devices **626**, for example, a keyboard, a microphone, or a touchpad. In one embodiment, the output device **625** also acts as the input device. An example of such a device may be a touch sensitive display. The input/output controller **624** may also output data to devices other than the output device, e.g., a locally connected printing device. In some embodiments, a user may provide input to the input device(s) **626** and/or receive output from the output device(s) **625**.

(48) The functionality described herein can be performed, at least in part, by one or more hardware logic components. According to an embodiment, the computing apparatus **618** is configured by the program code when executed by the processor **619** to execute the embodiments of the operations and functionality described. Alternatively, or in addition, the functionality described herein can be performed, at least in part, by one or more hardware logic components. For example, and without limitation, illustrative types of hardware logic components that can be used include Field-programmable Gate Arrays (FPGAs), Application-specific Integrated Circuits (ASICs), Program-specific Standard Products (ASSPs), System-on-a-chip systems (SOCs), Complex Programmable Logic Devices (CPLDs), Graphics Processing Units (GPUs).

(49) Although described in connection with an exemplary computing system environment, examples of the disclosure are operative with numerous other general purpose or special purpose computing system environments or configurations. Examples of well-known computing systems, environments, and/or configurations that may be suitable for use with aspects of the disclosure include, but are not limited to, mobile computing devices, personal computers, server computers, hand-held or laptop devices, multiprocessor systems, gaming consoles, microprocessor-based systems, set top boxes, programmable consumer electronics, mobile telephones, network PCs, minicomputers, mainframe computers, distributed computing environments that include any of the above systems or devices.

(50) Examples of the disclosure may be described in the general context of computer-executable instructions, such as program modules, executed by one or more computers or other devices. The computer-executable instructions may be organized into one or more computer-executable components or modules. Generally, program modules include, but are not limited to, routines, programs, objects, components, and data structures that perform particular tasks or implement particular abstract data types. Aspects of the disclosure may be implemented with any number and organization of such components or modules. For example, aspects of the disclosure are not limited to the specific computer-executable instructions or the specific components or modules illustrated in the figures and described herein. Other examples of the disclosure may include different computer-executable instructions or components having more or less functionality than illustrated and described herein.

(51) Aspects of the disclosure transform a general-purpose computer into a special purpose computing device when programmed to execute the instructions described herein. The detailed description provided above in connection with the appended drawings is intended as a description

of a number of embodiments and is not intended to represent the only forms in which the embodiments may be constructed, implemented, or utilized. Although these embodiments may be described and illustrated herein as being implemented in devices such as a server, computing devices, or the like, this is only an exemplary implementation and not a limitation. As those skilled in the art will appreciate, the present embodiments are suitable for application in a variety of different types of computing devices, for example, PCs, servers, laptop computers, tablet computers, etc.

(52) The term “computing device” and the like are used herein to refer to any device with processing capability such that it can execute instructions. Those skilled in the art will realize that such processing capabilities are incorporated into many different devices and therefore the terms “computer”, “server”, and “computing device” each may include PCs, servers, laptop computers, mobile telephones (including smart phones), tablet computers, and many other devices. Any range or device value given herein may be extended or altered without losing the effect sought, as will be apparent to the skilled person. Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.

(53) While no personally identifiable information is tracked by aspects of the disclosure, examples may have been described with reference to data monitored and/or collected from the users. In some examples, notice may be provided to the users of the collection of the data (e.g., via a dialog box or preference setting) and users are given the opportunity to give or deny consent for the monitoring and/or collection. The consent may take the form of opt-in consent or opt-out consent.

(54) The order of execution or performance of the operations in examples of the disclosure illustrated and described herein is not essential, unless otherwise specified. That is, the operations may be performed in any order, unless otherwise specified, and examples of the disclosure may include additional or fewer operations than those disclosed herein. For example, it is contemplated that executing or performing a particular operation before, contemporaneously with, or after another operation is within the scope of aspects of the disclosure. It will be understood that the benefits and advantages described above may relate to one embodiment or may relate to several embodiments. When introducing elements of aspects of the disclosure or the examples thereof, the articles “a,” “an,” and “the” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. The term “exemplary” is intended to mean “an example of.”

(55) Having described aspects of the disclosure in detail, it will be apparent that modifications and variations are possible without departing from the scope of aspects of the disclosure as defined in the appended claims. As various changes may be made in the above constructions, products, and methods without departing from the scope of aspects of the disclosure, it is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

Claims

1. A computerized method for delivering an application to a client machine in a third-party environment, the computerized method comprising: sending, by an application manager on the client machine to an application publishing portal associated with the third-party environment, a request to download the application from the application publishing portal, wherein an application plugin associated with the application is published by the application publishing portal, wherein the application is stored in an application repository; in response to the request to download the

application from the application publishing portal, receiving, from the application publishing portal, the application plugin associated with the application; executing, by the application manager, the application plugin associated with the application on the client machine, wherein the application plugin associated with the application comprises metadata and instructions to locate the application on the application repository; and using, by the application manager, the metadata and the instructions to cause the application to become available to a user of the client machine on-demand without downloading the application to the client machine from the application publishing portal or the application repository.

2. The computerized method of claim 1, wherein an application packager publishes the application plugin associated with the application on the application publishing portal, the computerized method further comprising: creating, by the application packager, the application plugin associated with the application in a format supported by the application publishing portal.

3. The computerized method of claim 1, wherein the application plugin associated with the application is automatically downloaded to the client machine upon detecting a login into the client machine by the user.

4. The computerized method of claim 3, wherein the application is available to the user of the client machine in real time upon detecting the login into the client machine by the user.

5. The computerized method of claim 1, wherein the application plugin associated with the application makes a procedure call to the application manager, and wherein the application manager locates the application in the application repository and makes the application available from the application repository to the client machine.

6. The computerized method of claim 1, further comprising: determining, by the application plugin associated with the application, that the application manager is not available on the client machine; and based on the determining, making, by the application plugin associated with the application, the application available on the client machine from the application repository without use of the application manager.

7. The computerized method of claim 1, further comprising: creating, by an application packager, the application plugin associated with the application.

8. The computerized method of claim 1, wherein the metadata describes the application including a location of the application in the application repository.

9. The computerized method of claim 1, wherein the metadata is in JavaScript Object Notation (JSON) format or Extensible Markup Language (XML) format.

10. The computerized method of claim 1, wherein the instructions in the application plugin associated with the application comprise executable instructions that trigger execution of the application manager on the client machine.

11. The computerized method of claim 1, wherein the instructions in the application plugin associated with the application comprise executable instructions that trigger a remote procedure call (RPC) end point in the application manager.

12. The computerized method of claim 11, wherein the application manager selects, based on the RPC end point, an application format from the application repository depending on an execution environment of the application plugin associated with the application on the client machine.

13. The computerized method of claim 1, wherein the application plugin associated with the application includes instructions to make the application unavailable to the user on the client machine.

14. A computer system for delivering an application to a client machine in a third-party environment, the computer system comprising: a processor; and a non-transitory computer storage medium having stored thereon program code executable by the processor, the program code causing the processor to: publish an application plugin associated with the application to an application publishing portal associated with the third-party environment; assign the application plugin associated with the application to a user of the client machine; store the application in an

application repository; receive, from the client machine, a request to download the application from the application publishing portal; and in response to the request to download the application from the application publishing portal, provide the application plugin associated with the application to the client machine, wherein the application plugin associated with the application comprises metadata and instructions to locate the application on the application repository, and wherein an application manager executes the application plugin associated with the application and uses the metadata and the instructions to cause the application to become available to the user of the client machine on-demand without downloading the application to the client machine from the application publishing portal or the application repository.

15. The computer system of claim 14, wherein the program code further causes the processor to create the application plugin associated with the application in a format required by the application publishing portal.

16. The computer system of claim 14, wherein the application is made available to the user of the client machine in real time upon detecting a login into the client machine by the user.

17. The computer system of claim 14, wherein the application plugin associated with the application is a fraction of a size of the application.

18. A non-transitory computer storage medium having stored thereon program code executable by a processor, the program code embodying a program code method for delivering an application to a client machine in a third-party environment comprising: publishing an application plugin associated with the application to an application publishing portal associated with the third-party environment; assigning the application plugin associated with the application to a user of the client machine; storing the application in an application repository; receiving, from the client machine, a request to download the application from the application publishing portal; and in response to the request to download the application from the application publishing portal, providing the application plugin associated with the application to the client machine, wherein the application plugin associated with the application comprises metadata and instructions to locate the application on the application repository, and wherein an application manager executes the application plugin associated with the application and uses the metadata and the instructions to cause the application to become available to the user of the client machine on-demand without downloading the application to the client machine from the application publishing portal or the application repository.

19. The non-transitory computer storage medium of claim 18, wherein the program code method further comprises creating the application plugin associated with the application in a format required by the application publishing portal.

20. The non-transitory computer storage medium of claim 19, wherein the application plugin associated with the application is not a portion of the application.
